MICROPROCESSORS

Course Objective:

- To understand the Assembly language programming using 8085 microprocessor instruction set.
- To understand the concept of interfacing of 8085 microprocessor with different ICs.

Course Outcomes

Student should be able to:

- Learn internal organization of some popular microprocessors.
- Understand the hardware and software interaction and integration of different microprocessors.
- Implement the 8085 programming for different field applications.
- Understand the basic idea about data transfer schemes and its applications.

Unit-1

Introduction to microcomputer, CPU, microprocessors (8085, Z-80, Motorola 6800 CPU), General 8-bit microprocessors, Architecture of 8085 microprocessor and its functional blocks. ALU, Timing and control unit, Interrupts, flag register, general purpose registers, PC and SP, and different pins.

Unit-2

Instruction set of 8085 CPU- Data transfer group; Arithmetic group; Logic group; Branching group; stack operation, I/O and Machine control group.
Memory and I/O interfacing, Interfacing of 8085 with 64K x8, 16K X8, 8K X8, 4K X8 bit memory RAM/ROM chips. Consideration of loading effect.
Simple assembly language programming practices on data transfer, arithmetic, logic, stack and subroutines, I/O, etc.

Unit-3

T- state, Machine cycle, Instruction cycle, fetch and execution operations, timing diagrams, Estimation of execution time. Different data transfer modes, PPI 8255, USART 8251; Architecture of PPI 8255 and its functional blocks; I/O ports; programming of 8255 in I/O and BSR mode; application of 8255 in different I/O modes. Architecture of USART 8251, and its programming in different modes. Idea of the use of 8279, 8259 chips.

Unit-4

Data transfer schemes, programmed I/O, interrupt structure of 8085, and interrupt driven I/O, interfacing of A/D and D/A converters, Data acquisition systems, temperature control, waveform generation and stepper motor control.

Textbooks:

1. R. S. Gaonkar, "Microprocessor Architecture: Programming and Applications with 8085", Penram International Publishing, 1996.
2. Ghosh and Shridhar, "0000 to 8085 Microprocessor".

References:

D. V. Hall, "Microprocessors and Interfacing", Mc Graw Hill Higher Education, 1991.